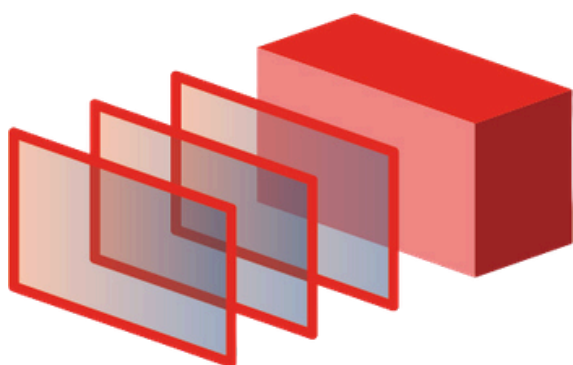


FIAMM
+ -

ENGINEERING MANUAL



VALVE REGULATED
LEAD ACID BATTERIES
(VRLA BATTERIES)

AGM TECHNOLOGY

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INTRODUCTION

In a high technological environment, it is extremely important to have a backup power source whenever possible. In fact, mains power failure could cause severe losses and damages anytime.

FIAMM ENERGY TECHNOLOGY (FET) has been developing throughout years of research and experience several ranges of AGM (Absorbed Glass Material) lead acid batteries to ensure the best reliability and quality.

1. CONSTRUCTION FEATURES

The main construction features of FET AGM batteries are shortly described in the below section.

1.1 Plates

Both positive and negative plates (A) are of the flat pasted type. The active material is made of a paste of lead oxide, water, sulfuric acid and other materials needed to obtain the performance and stability required throughout battery life. The grids are made of a high-quality lead alloy with calcium and tin which assures good resistance against corrosion.

1.2 Containers

Battery cases and lids (B) are made of a type of ABS which complies with American Standards UL 94 (for specific models, flame retardancy class V-0 according to IEC 60695-11-10, method FV-0). This material is shock-resistant and flame-retardant. They are also designed to fully withstand the internal pressure variations during battery operation. This is further ensured by reinforced container walls and lids. Handles have been designed for some batteries into the lids to facilitate handling.

1.3 Separators

The separators (C) are made of glass microfiber mats by a special process which results in a high porosity with very small pore diameters to ensure maximum oxygen diffusion while maintaining high plate utilization and low internal resistance. The plates are completely wrapped by the separator, and the electrolyte is completely absorbed in the separator and plates. By this method, the shedding of active material, which during the battery life causes short circuits in flooded battery constructions, is avoided.

1.4 Electrolyte

The electrolyte is sulfuric acid of 1,3 sp. gr. at 20°C with same purity characteristics as other types of high-quality lead acid batteries.

1.5 Valves

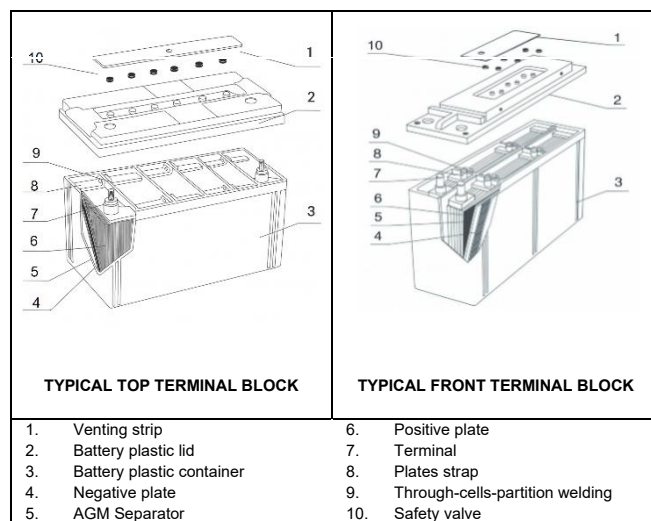
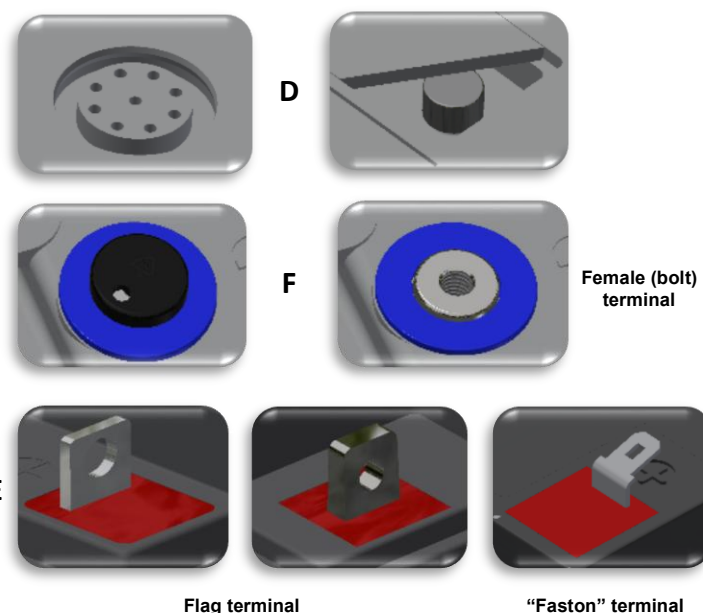
Each cell has a one-way valve (D) to permit the release of gases from the cell whenever the internal pressure exceeds the fixed safety value. The valve is rated at 0,15~0,30 atmospheres (15~30 kPa).

1.6 Terminal posts

Suitable threaded post designs (E) ('female' or 'faston' pole) with solid connectors are provided to ensure low ohmic losses. Posts to lid seals are designed to prevent leakage over a wide range of internal pressures and conditions of thermal cycling. Intercell connections in the FET multicell AGM battery design is electrically welded through the cell walls to minimize the internal impedance while maintaining complete separation of the individual cells.

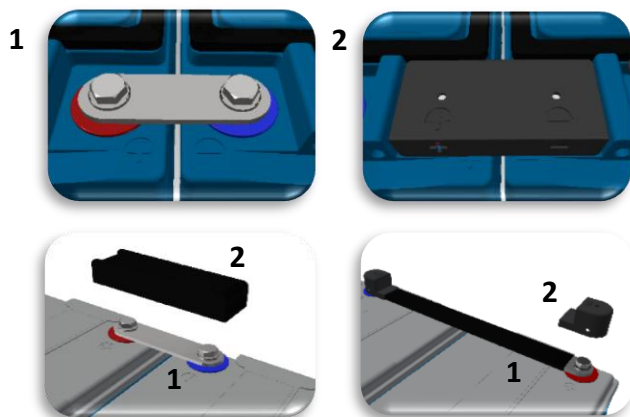
Special plastic terminal caps (F) are provided for transportation assuring a protection against short circuits during transportation.

Cutaway drawing of FET AGM cell



1.7 Connections (where supplied)

Suitable solid connectors (1) are made of tin plated copper and have a length from 5 to 10 mm; suitable insulated plastic covers (2) are made of ABS -V0.



TOP TERMINAL CONNECTION is mainly used for FET battery range SLA – FLB as well.

1.8 Front Terminal Connections

Suitable solid connectors (1) made of tin plated copper and covers (2) made of ABS are provided to make a proper installation between blocks. Designed for FET front terminal range FIT using a special “L” clamp (3).



1.9 Remote Venting System (RVS)

Most of FET AGM batteries are designed with an optional central degassing system on the top lid covering sheet. So, if batteries need to be installed in a totally sealed cabinet, it is advisable to use the remote venting system available from the manufacturer to conduct any gas from the batteries to the outside of the cabinet itself.



RVS – FRONT side
used for FIT Range



RVS – TOP side
used for SLA / FLB / FIT
Range

2. OPERATING FEATURES

2.1 Capacity

The battery capacity is rated in ampere-hours (Ah) and is the quantity of electricity which it can supply during discharge. The capacity depends on quantity of active materials contained in the battery (thus on dimensions and weight) as well as the discharge rate, and temperature, and minimum voltage.

The nominal capacity of FET batteries refers to the 10 hours discharge rate (indicated with C_{10}) with constant current at 20°C to 1,80 volt per cell.

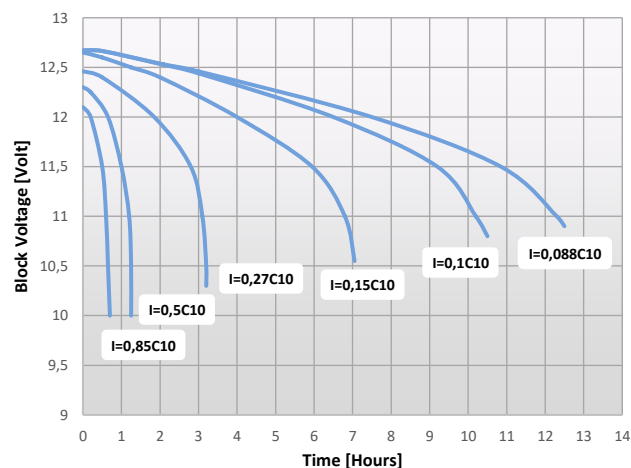


Fig. 1:
Typical discharge curves for FET AGM batteries (FIT range).

2.2 Capacity in relation to discharge rate

The available capacity of all lead acid batteries depends on discharge rate (discharge current); this is due to internal electrochemical process and type of construction (i.e. type of positive plate).

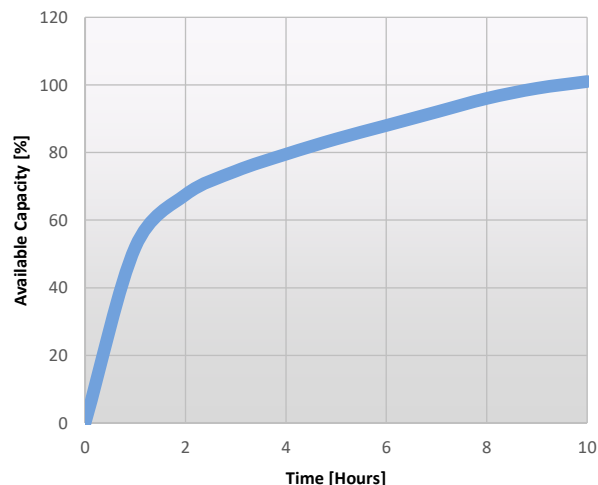


Fig. 2:
Average available capacity versus discharge rates for FET AGM batteries.

2.3 Capacity range of FET AGM Lead Acid Batteries

FET Battery range	Capacity Range 10h 1,80 VPC 20°C [Ah]
FG	from 1,1 to 37,9
FGH	from 4,6 to 16,5
FGHL	from 4,6 to 15
FGL	from 15 to 184
SP	from 22,2 to 223
FIT	from 40 to 195
FHT ²	from 96 to 180
SLA	from 50 to 2000
FLB	from 23 to 194
FHB	from 87,3 to 194

2.4 Capacity in relation to the temperature

The capacity available from a battery, at any discharge rate, varies with temperature.

Batteries which have to operate at temperatures different from the nominal (20°C) need a higher or lower capacity as per the factor indicated in the following graph (required capacity at 20°C is selected dividing the capacity at temperature "X" by the correction factor stated in the graph).

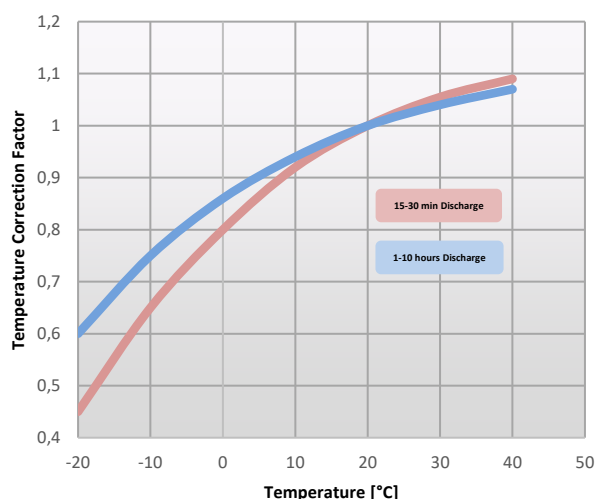


Fig. 3:
Capacity Correction factor versus temperature for a 10 hours discharge rate for FET Lead Acid AGM batteries.

2.5 Internal impedance and short circuit current

The internal impedance of a lead acid battery is a direct result of the type of internal construction, plate thickness, number of plates, separator material, electrolyte sp. gr., temperature and state of charge.

The internal resistance and the short circuit current of FET VRLA batteries at 100% state of charge and 20°C is indicated in the relative Product Sheet. These values are calculated in accordance with IEC 60896-part 21/22.

Different instruments are available to detect the internal resistance or impedance of lead acid batteries. These instruments use a different way to determinate these values. The values obtained from these instruments will be different to the values stated in FET Product Sheet.

2.6 Service life

According to the main international standards a battery is considered at the end of its service life whenever delivering less than 80% of its nominal capacity. The recommended operating temperature range is between 10°C to 30°C (FHT² range peaks up to 45°C).

FET VRLA batteries can operate over a temperature range of -20 to +50°C and higher; operation at temperature higher than 20°C reduces life expectancy according to the graph in figure 4.

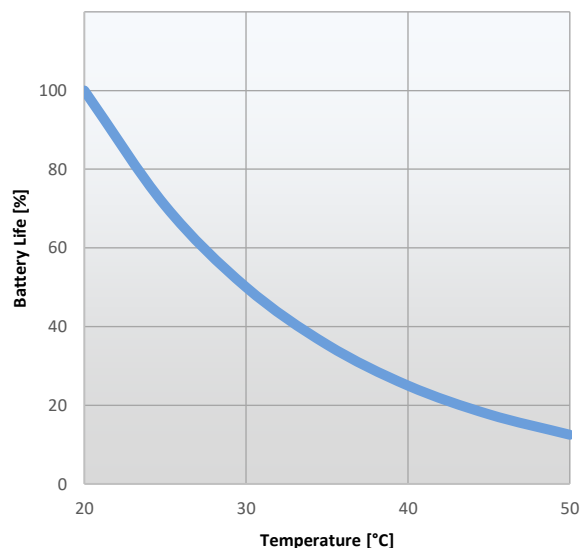


Fig. 4:
Expected service life vs. working temperature.

2.7 Gassing

All Lead Acid Batteries emit gases during the charge process. FET VRLA batteries have a high recombination efficiency (>98%) and for cells operated at 20°C under normal operating conditions venting is virtually negligible. Laboratory test measurements show the following gassing rates:

- 2 ml/Ah/cell/month at a float voltage of 2,27 V/cell.
- 25 ml/Ah/cell/month at a boost voltage of 2,40 V/cell.

The quantity of gas given off in the air (it basically consists of 80-90% hydrogen) is very low and thus it is clear that FET VRLA batteries can be installed in rooms containing electric equipment with no explosion danger or corrosion problems under normal conditions. In any case these rooms or cabinets must have natural or forced ventilation and not be fully sealed. Please refer to chapter 11 "Ventilation" for information on required air exchange.

2.8 Operation of batteries in parallel

When the required capacity exceeds the capacity of a single string of batteries, it is possible to connect more strings in parallel paying attention to the following guidelines:

- In each string only cells or monoblocks of the same type, model, manufacturing date and quantity should be used.
- A symmetrical layout of the batteries should be designed (i.e. length and type of connector) to minimize possible resistance variations.
- The quantity of strings in parallel should be reasonable in terms of layout and application. Usually, 4 strings could be connected in parallel. However, depending on strings voltage and cables length, a higher number of strings could be safely connected to reach required total capacity.

2.9 Open circuit voltage - State of charge

The measurement of the open circuit voltage (battery has to be disconnected from charger system for at least 24 hours) provides an approximate indication of the state of charge of the cells.

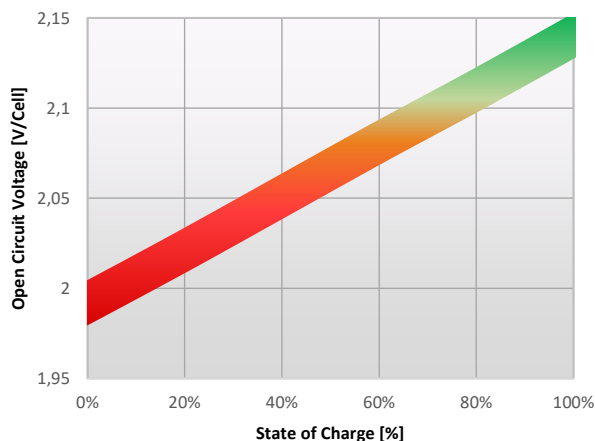


Fig. 5:
Approximate state of Charge versus Open circuit cell voltage.

2.10 Applications

The table below summarizes the main (■) and secondary (□) applications for each battery range.

	FG	FGH	FGHL	FGL	SP	FIT	FHT ²	SLA	FLB	FHB
Data Center & UPS	■	■	□	■	□	■	□	■	■	■
Telecom						■	■	■	□	□
Industry & Utilities						■	□	■	□	
Motion			□	□	■	■	■	■	□	□
Security & Emergency	■		■	■				□		
Renewable			■	■						

3. CHARGING

In order to ensure the best protection against power failures in any moment, it is necessary that batteries are kept in the following conditions:

- In float charging throughout all their standby period.
- Fully recharged soon after a discharge.
- Completely recharged after a discharge. Recharge as soon as possible to ensure maximum protection against subsequent power outages. Early recharge also ensures the maximum battery life.

3.1 Floating charge

Floating systems have the charger, battery, and load connected in parallel. The float setting keeps the battery fully charged with minimal water consumption.

The recommended float voltage is 2,27 V per cell at 20°C. The optimal float voltages to maximize the battery life over the range of temperatures between -20 and +50°C are shown in the figure 6 or using the formula: **-2,5mV/cell/°C**.

In fully charged FET AGM batteries at 2,27 VPC and 20°C, the typical float current is about 0,3 mA/Ah. Because of the nature of recombination phenomena, the float current observed in FET AGM batteries is normally higher than in vented batteries and is not an indication of the state of charge of batteries.

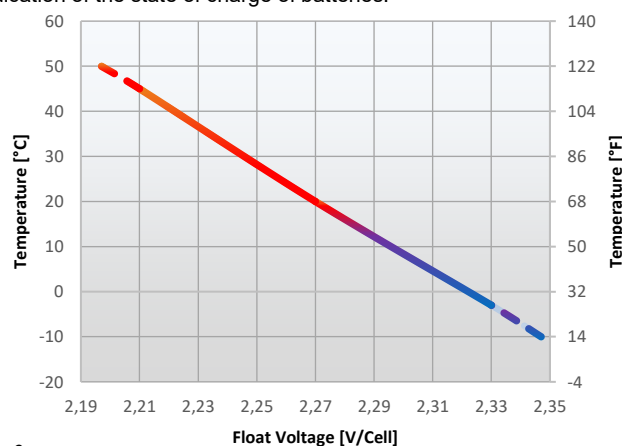


Fig. 6:
Recommended float voltage at different temperatures.

3.2 Boost charge

Boost charge should be performed periodically (every 12 months), mainly to prevent battery stratification. Use a constant voltage of 2,40 V/cell at 20 °C with a maximum current limit of 0,25 C₁₀. The recharge should be stopped when the current through the battery drops below 0,03 C₁₀, or when the battery temperature rises more than 10 °C above the ambient temperature. For temperature compensation, follow the floating charge criteria.

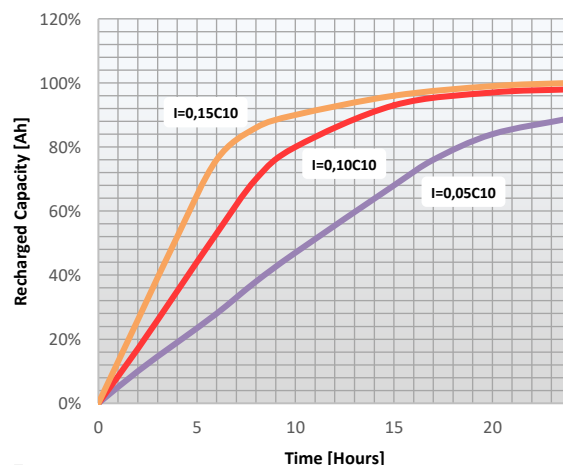


Fig. 7:
Boost charge curves at 2,40 V/cell with different limits of current.

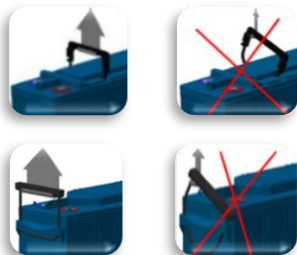
4. BATTERY INSTALLATION

All necessary precautions must be taken when working with lead acid batteries as per electrical risk, explosives gasses, heavy components, corrosive liquids. Use insulated tools and wear protective equipment.

4.1 Installation

FET VRLA batteries can be fitted on stands or into cabinets. FET offers a wide selection of stands, from one tier/one row to six tiers/three rows, to suit most applications. Cabinets are available with or without circuit breaker and its relevant compartment.

1. Avoid any impact or shock which could cause breaking or micro fractures in the container. Do not lift cells by its terminals. Always lift the individual unit from underneath, or by the build-in-lifting handles.



2. Ensure all cell jars and covers are completely clean and dry.
3. Synthetic cleaning cloths must not be used. Clean lids and containers only with antistatic cotton cloths soaked in a solution of mild soap and completely "wrung" out.
4. Should the terminal posts have a white film on them, lightly abrade their contact surfaces, using a Scotchbrite pad or fine grit abrasive paper, to remove any surface oxidation.
5. Do not lift cells by terminals; do not use terminals as point of lifting during handling/installation process.
6. Place the single units at their correct position according to the electrical layout.
7. In order to allow heat dissipation, a minimum distance of **5–10 mm** between cells/blocks is recommended in accordance with IEC 62485-2; this is normal when using FET standard connectors; for special requirements, please contact FET.
8. Care must be taken to avoid short circuiting the cells with any of the battery hardware.
9. Start with the lowest shelf to ensure stability. Carefully preserve the sequence: positive, negative, positive, negative throughout the whole battery. Flexible cable connectors for connecting from one shelf to the one below, will be applied once all the blocks have been connected (we would suggest connecting such inter-shelf or inter-row cable connectors at the final User's premises only).
10. To ensure a good electrical contact between the bottom of each terminal and the connecting strap and, at the same time, to ensure that the threaded terminals are not damaged by excessive torque, use a torque spanner set on the value of:

FIAMM Battery Range	TYPE	VALUE [Nm]
FLB-FGL	M5	5÷6
FLB-FGL-SP-FIT-SLA	M6	7÷9
FLB-FGL-SP-FIT-SLA-FHT²	M8	10÷12
SLA	M10	20÷25
FG-FGH	Flag Ø5,5	5÷6
FG	Flag Ø6,5	6÷7

11. Insulate all the connectors by means of the plastic covers being supplied with the battery accessories (if applicable).
12. Affix the cell number stickers to the cell jars making sure that the surfaces are dry and clean. It is usual to number the cells beginning with #1 at the positive end of the battery, numbering

consecutively in the same order as the cells are connected electrically, through to the negative end of the battery.

13. Check the total battery voltage which should comply with the total number of cells connected in series.
14. The cells are usually designed to be installed in vertical position; the horizontal position could in some cases stress the cells. Here below the correct cell arrangements for large capacity AGM cells.

HORIZONTAL HANDLING



If the cells (SLAH) have to be placed horizontally on a rack, we suggest using a special strap (Fig. 8).

Fig. 8:
Image of strap for horizontal handling for SLAH.

HORIZONTAL INSTALLATION



Example of installation and cell arrangement (position of + / - poles) of AGM batteries with capacity equal or higher than 800Ah.

Fig. 9:
Image of horizontal installation.

HANDLE



Do not handle the cells from the poles.

Fig. 10:
Dangerous handle the cells from the poles.

4.2 Batteries installed into cabinet

For safety reasons, we would not recommend to pre-assemble the blocks into the cabinets before shipment to the final Customer. However, if this is normal practice for some system makers, we would strongly recommend paying special attention to protect the battery system from mechanical stress and vibrations occurring during transport. For this purpose, we would require to properly fasten all the blocks to the relevant cabinet shelves by means of plastic band and/or other adequate methods. Furthermore, the cabinet should be protected, in the outside, with shock-absorbing packaging material, in order to prevent any transmission of vibrations to the internal components such as the battery blocks. Special precautions must be taken to avoid accidental short circuits (do not connect all the batteries, divide the battery circuit low voltage parts).

For any further information please refer to IEC 62485-2 Standard or contact FET at: info.standby@fiamm.com

4.3 Ripple

Voltage ripple: Residual AC ripple is usually present in the output voltage of chargers; amplitude and frequency depend on charger design, and it can affect negatively battery life. Ripple could increase water loss, battery temperature and accelerate corrosion with a result to reduce the battery life. It is recommended therefore, that voltage regulation across the system including the load, but without the battery connected, under steady state conditions, shall be better than $\pm 1\%$ through 5% to 100% load. Transient and other ripple type excursions can be accommodated provided that, with the battery disconnected but the load connected, the system peak to peak voltage, including the regulation limits, falls within 2,5% of the recommended float voltage of the battery.

Current ripple: under no circumstances should the current flowing through the battery when it is operating under float conditions, reverse into the discharge mode.

4.4 External Devices on Battery System

In case devices of any kind are connected to the batteries and draw energy from them, the following conditions must be strictly observed:

1. The current draw must be balanced, ensuring that energy is evenly drawn from each battery block.
2. The battery power supply must never be turned off. If it is necessary to interrupt the power supply for more than a couple of days — and not in a cyclical or temporary way — all devices consuming energy must be disconnected, in order to prevent deep discharges or damage to the system.

4.5 Battery room requirements

- The battery room should be dry, well-ventilated and have its temperature as moderate as the climate will allow, preferably between 10°C and 30°C.
- **NO smoking or open flames in the battery room.**
- Adequate ventilation to change the air in the battery room is essential to prevent an accumulation of the gases given off during charge (for further information please refer to chapter 11 "Ventilation").
- The battery will give the best results and life when working in a room temperature of 20°C (FHT² - FHB range can satisfactorily operate at higher temperatures). High temperatures increase the performance but decrease the life of the cells; low temperatures reduce the performance.
- Keep the battery out of direct sunlight.
- If a rack is not supplied by FET, suitable racks should be provided to support the cells. These should be arranged to provide easy access to each cell for inspection and general maintenance. Suitable racks may be made of wood or metal with a coating of acid-resistant paint. If metal racks are used, they must be fitted with rubber or plastic insulators to prevent the cells meeting the metal.
- To facilitate proper battery operation, maintenance, and care, post a battery data card/ instruction table in a conspicuous place near the battery to provide the attendant with service information and data.

5. SAFETY

It is always recommended that full precautions be taken when working on batteries. The safety standards of the country of installation must be risk, explosives gasses, heavy components.

5.1 Protective Equipment

Make sure that the following equipment is available to personnel working with batteries:

- Instructions manual.
- Tools with insulated handles.
- Fire extinguisher.
- PPE (Personal Protective Equipment) must be worn (glasses, gloves, aprons, etc.). To avoid static electricity when handling batteries, material of clothing, safety boots and gloves are required to have a surface resistance $\leq 10^8 \Omega$, and an insulation resistance $\geq 10^5 \Omega$.
- First aid equipment must be available.

5.2 Safety Precautions

Always observe the following precautions:

- Batteries are no more dangerous than any other equipment when handled correctly.
- Do not allow metal objects to rest on the battery or fall across the terminals (even when disconnected, a battery remains charged!).
- Never wear rings or metal wrist bands when working on batteries.
- Please be aware that fast discharge rates (i.e. 15-minutes discharge rates) may lead to a significant temperature increase of batteries connections so take care to avoid coming into contact with those hot surfaces. 
- Do not smoke or permit open flames near batteries or do anything to cause sparks.
- Do not try to remove the battery cap to add water or acid into the cell(s).
- Never lift or pull up the battery at the terminals.
- Air exchange must be provided to prevent the formation of explosive hydrogen concentration.
- For further information please refer to IEC 62485-2 Safety requirements for secondary batteries and battery installations Part 2: Stationary batteries.

5.3 Battery Disposal

Lead acid batteries must be disposed according to the country's law. It is strongly recommended to send batteries for recycling to a lead smelter. Please refer to the local Standards for any further information; these batteries need to be collected separately for waste disposal. As of the 31st December 1994, all Valve Regulated Lead Acid (VRLA) battery has to have the following symbols present in conformance to EG-guideline 93/86/EEG.



Pb

6. MAINTENANCE

6.1 Battery care

GASES GIVEN OFF BY BATTERIES ON CHARGE ARE EXPLOSIVE!

DO NOT SMOKE OR PERMIT OPEN FLAMES OR DO ANYTHING TO CAUSE SPARKS NEAR BATTERIES.

1. Keep the battery and surroundings clean and dry.
2. Make sure that bolted connections are properly tightened (chapter 4 "Battery Installation").
3. Usually, it is not necessary to apply grease on the bolts and connectors, in any case "No-oxide" grease increase the protection against corrosion.
4. Should any corrosion of the connections occur because of spilled acid, etc., carefully remove corrosion materials, thoroughly clean and neutralize with diluted ammonia or baking soda.
5. Keep the battery at the recommended charge voltage (see chapter 3 "Charging").
6. The room in which the battery is installed should be well ventilated and its temperature as close as possible to 20°C.
7. Do not try to open the cover valve.

6.2 Cleaning

When necessary, batteries could be cleaned using a soft dry antistatic cloth or water-moistened soft antistatic cloth paying attention not to cause any ground faults.

No detergent or solvent-based cleaning agents or abrasive cleaners should be used as they may cause a permanent damage to the battery plastic container and lid.

6.3 Voltage checks

All voltage measurements should be made when the whole battery has stabilized on floating, at least 7 days after battery installation or after a discharge/charge cycle.

To facilitate voltage reading in the correspondence of each block terminal protection covers are designed with a safe and proper hole. Measure and record individual block voltages on float once a year. The blocks/cells should be within $\pm 4\%$ of the average at 20°C. No corrective action is required in this case.

Maintaining a correct battery charging voltage is extremely important for the reliability and life of the battery. So, it is advisable to carry out a periodical checking of the overall float voltage to verify any possible defect of charger or connections.

6.4 Cell appearance

Any cells showing corrosion, container bulging, high temperature than the other cells, should be regarded as suspect. Such cells should be carefully examined and expert advice should be obtained immediately from FET.

6.5 Pilot Cell

For regular monitoring of the battery condition, select one or more battery cells as a "pilot" cell(s); for batteries comprising more than 60 cells, select one pilot cell for every 60 cells.

6.6 Maintenance / Periodic Inspections

Written records must be kept of battery maintenance, so that long-term changes in battery condition may be monitored.

The following inspection procedures / maintenance activities are recommended:

EVERY SIX MONTHS:

- Visual inspection on cells/racks (appearance, cracks or corrosion signs, electrolyte leakage...).
- Check and record the overall float voltage at the battery terminals (not at the charger!).
- Measure and record cell voltage (on minimum 3% of total cells randomly).
- Measure and record the pilot cell(s) temperature.
- Room ventilation.

YEARLY:

- All the controls indicated at six months.
- Check and record the voltage of all cells.
- Clean the cells.
- Discharge test followed by boost charge.

YEARLY SUGGESTED:

- Make sure all connections are torqued according to connection torque table; in case of frequent high discharge current please consider checking.

7. APPLICABLE STANDARDS

FET Valve Regulated Lead Acid Batteries comply with:

- **IEC 60896 – 21:**
Stationary lead-acid battery Valve Regulated Type – Part 21: Valve regulated types - Methods of test.
- **IEC 60896 – 22:**
Stationary lead-acid battery Valve Regulated Type – Part 22 S: Valve regulated types - Requirements.
- **IEC 62485 – 2:**
Safety requirements for secondary batteries and battery installations Part 2: Stationary batteries.
- **IEC 60695 – 11 – 10:**
Fire hazard testing - Part 11 – 10: Test flames - 50 W horizontal and vertical flame test methods.
- **BS 6290 – 4:**
Lead-acid stationary cells and batteries. Specification for classifying valve regulated types.
- **BS 6290 – 1:**
Lead-acid stationary cells and batteries. Specification for general requirements.

8. BATTERY TEST

As prescribed by EN 60896-21/22, any discharge test must be conducted with a battery in a fully charged condition.

To achieve this condition it might be necessary to perform a boost charge (2,40 V per cell at 20 °C, blocking it when the current is below 0,03 C₁₀).

In order to take temperature readings of a battery, one pilot cell or block shall be chosen. The surface temperature of the container wall centre of each pilot cell or block shall be measured immediately prior to the discharge test.

The individual readings shall be between 15°C and 30°C. The temperature of the selected block shall be considered as representative of the average temperature of the battery.

It is desirable that the average cell surface temperature and the ambient temperature fall as nearer to the reference temperature of 20°C or 25°C as possible.

In case of batteries having a capacity lower than 80% of the nominal rating it is advisable to replace them within 12 months.

Here below some precautions to be taken:

- Discharge must be stopped at the final discharge voltage.
- Deeper discharges must not be carried out unless specifically agreed with FET.
- Recharge immediately the battery after each (full or partial) discharge test.

8.1 Service/Functional test

This is a test of the battery's ability to satisfy the design requirements of the system. It means to discharge the battery directly to the load (in this case take precautions to ensure that a battery failure does not jeopardize other equipment) or dummy load to simulate a main failure.

1. Record the floating voltage of each cell, as well as the total system voltage.
2. Check the actual load (A or W), as well as the minimum admissible voltage of the system.
3. On FET discharge tables you can approx. determine the discharge rate (minutes of discharge). **Please note that battery performances decrease with battery age.** After switching off the rectifier, discharge the battery for a time of 20% of that calculated discharge rate.
4. During the discharge, record at regular intervals, cell/block voltage, battery temperature, discharge current, total battery voltage.
5. For safety reasons, during the test assure that the total battery voltage remains above the minimum depending on discharging rate in order to avoid any failure to the system (please note that approaching to the final voltage, the voltage curve decreases rapidly).
6. Wait at least 48 hours between two consecutive discharge tests (in the event of a full discharge).
7. For comments on test data, please refer to FET technical offices.

8.2 Capacity test

Please carry out this test only when complete information on the quantity of energy inside the battery is requested.

Take precautions because after this test battery SHOULD NOT BE ABLE TO SUPPLY ENERGY IN CASE OF MAIN FAILURE.

Dummy load is usually necessary to provide the request discharge current. Test is usually carried out to verify the battery capacity to a specific end voltage and discharge rate (usually 1, 3 or 10 hours).

Test must be conducted in accordance with EN 60896-21/22. Please refer to the prescription indicated in the above standard.

Record at regular intervals every half an hour at the beginning, every 10 minutes the last half an hour cell/block voltage, battery temperature on pilot cell, discharge current, total battery voltage (in any case voltage reading has to be made at least at 25%, 50% and 80% of the discharge time).

According to IEC60896-21 the discharge shall be terminated when one of the following values t_{disch} , whichever comes first, has been recorded:

1. t_{disch} = the elapsed time of discharge of the string, with n cells, to a voltage of $n \times U_{final}$ (V)
2. t_{disch} = the elapsed time when the first of the unit in the string reached a voltage of

$$U = U_{final} - \left(\sqrt{\frac{Unit\ voltage}{2}} \right) \times 0,2$$

At the end of discharging test, batteries must be recharged immediately (according to paragraph 3.2 "Boost charge").

The following formula determinates the battery capacity:

$$C = \text{discharge current} \times t_{disch}$$

(where t_{disch} is indicated in hours)

For temperatures different from the nominal (20°C) and discharge rates between 3 to 10 hours, the battery capacity shall be corrected as follows:

$$C_{20^{\circ}C} = \frac{C}{1 + \lambda(\theta - 20)}$$

Where:

θ = initial pilot cell temperature (°C)

$\lambda = 0,006$ for tests > 1 hour

$\lambda = 0,01$ for tests ≤ 1 hour

Trending battery capacity during years will provide information in predicting when the battery will no longer meet design requirements.

9. UNPACKING

9.1 Inspection

Upon receiving a shipment of battery cells, it is advisable to open the shipping containers and carefully check the cells and hardware against the packing list.

The contents of each consignment are carefully inspected by FET before shipment. Any damage must be reported immediately to the carrier and the damaged items retained for inspection by the carrier's representative.

9.2 Handling

AGM batteries units are shipped fully charged and must always be treated with care. The product can supply high short circuit currents, even if the case or lid is damaged. Always lift the individual unit from underneath, or by the build-in-lifting handles (please also refer to chapter 4 "Battery Installation").

Never apply force to, or drop anything on, the terminal posts: doing so may damage the threads or the post seals.

10. STORAGE

10.1 Storage prior to installation

VRLA batteries range are delivered activated/filled and charged ready for installation. If they cannot be installed immediately, the following instructions need to be respected.

10.2 Storage Conditions

A good storage practice requires as follows:

- Battery storage area must be clean, cool and dry.
- Surroundings must be kept clean.
- Elevated temperatures, direct and indirect sunshine must be avoided.
- Optimum storage temperature range is -10°C to +30°C.
- Avoid storage in ambient with a relative humidity greater than 90%.
- Battery cells must be protected from harsh weather, moisture and flooding.
- Storage on a pallet wrapped in plastic material is permitted, in principle. However, it is not recommended in rooms where temperature fluctuates significantly, or if high relative humidity can cause condensation under the plastic cover.
- With time, this condensation can cause a whitish hydration on the poles and lead to high self-discharge by leakage current.
- Atmospheres with chemical contaminants have to be avoided.
- Do not load other merchandise on top of unprotected batteries.
- Battery cells must be protected from dropping objects, from falling and falling over.
- Battery cells must be protected from short circuits by metallic parts or conductive contaminations.
- Avoid storing of unpacked battery cells on sharp-edged supports.
- Stacking of pallets is not permitted unless otherwise specified.
- It is recommended to realize the same storage conditions within a batch, pallet or room.

10.3 Storage time / Temperature

VRLA battery ranges have a shelf life of **9 months** at a storage temperature of **20°C**.

The temperature has an impact on the self-discharge rate of battery cells.

Higher temperatures increase the rate of self-discharge and therefore storage life is reduced.

MAXIMUM storage period before refresh at the given average storage ambient temperature is as follows:

Max storage time	Temperature
9 months	20 °C
6 months	30 °C
3 months	40 °C

10.4 Storage / Recharge

As during storage batteries will lose part of their capacity due to self-discharge, a refreshing charge must be given:

1. when MAXIMUM STORAGE TIME is reached

OR

2. when the OCV (open circuit voltage) approaches 2,09 V/cell

whichever occurs first.

Recharge the cells as directed in FET'S instruction table for AGM battery type (usually at 2,40 V per cell).

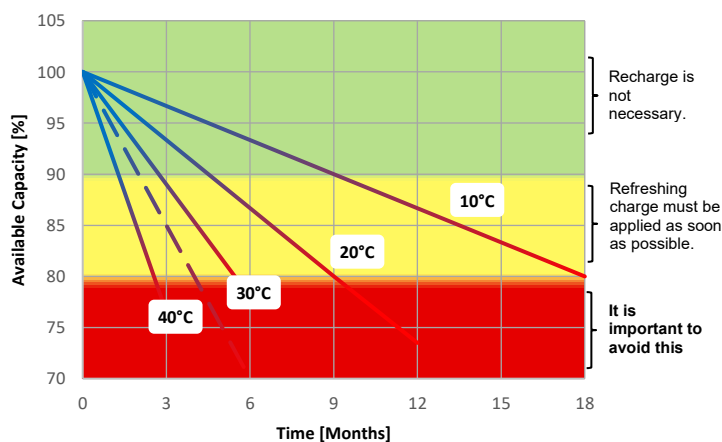


Fig. 11:
Storage time versus temperatures.

11. VENTILATION (in accordance with IEC 62485-2)

During normal operating conditions, lead acid batteries emit low quantity of gases which can reach an explosive mixture when hydrogen concentration is higher than Lower Explosion Limit (LEL) threshold which is 4%vol.

The purpose of ventilating a battery location or enclosure by natural or forced (artificial) ventilation is to maintain the hydrogen concentration below the above stated limit. Battery locations and enclosures are to be considered as safe from explosions, when the concentration of hydrogen is kept below this safe limit.

The minimum air flow rate for ventilation of a battery location or compartment shall be in accordance with European Standard IEC 62485-2 calculated by the following formula:

$$Q = 0,05 \times N \times I_{gas} \times C_{rt} \times 10^{-3}$$

where:

Q = ventilation air flow in m³/h

N = number of cells (each 2 volt)

C_{rt} = capacity C₁₀ [Ah] at 1,80 volt/cell at 20°C

The current I_{gas} [mA/Ah] producing gas as indicated in the table of the above-mentioned standard can be assumed as:

I _{gas} = 1	For batteries on float charge
I _{gas} = 8	For batteries on boost charge

11.1 Determination of openings

The amount of ventilation air flow shall preferably be ensured by natural ventilation, otherwise by forced (artificial) ventilation. Battery rooms or enclosures require an air inlet and an air outlet with a minimum free area of opening calculated by the following formula:

$$A = 28 \times Q$$

with Q = ventilation flow rate of fresh air [m³/h]

A = free area of opening in air inlet and outlet [cm²]

Note: For this calculation the air velocity is assumed to be 0,1 m/s.

The air inlet and outlet shall be located at the best possible location to create best conditions for exchange of air, i.e.

- Openings on opposite walls.
- Minimum separation distance of 2 m when openings on the same wall.

The following picture gives an indication of the correct opening to assure a complete battery room air exchange

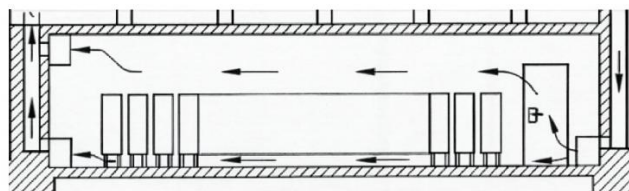


Fig. 12:
Opening on opposite walls.

11.2 Forced ventilation

Where an adequate air flow Q cannot be obtained by natural ventilation and forced ventilation is applied, the charger shall be interlocked with the ventilation system, or an alarm shall be actuated to secure the required air flow for the mode of charging selected. The air extracted from the battery room shall be exhausted to the atmosphere outside the building.

11.3 Close vicinity to the battery

In the close vicinity of the battery the dilution of explosive gases is not always secured. Therefore a safety distance extending through air must be observed within which sparking or glowing devices (max. surface temperature 300 °C) are prohibited. The dispersion of explosive gas depends on the gas release rate and the ventilation close to the source of release. For calculation of the safety distance d from the source of release the following formula applies assuming a hemispherical dispersal of gas. The safety distance "d" is given from the following formula:

$$d = 28,8 \times \sqrt[3]{N} \times \sqrt[3]{I_{gas} \times C_{rt}}$$

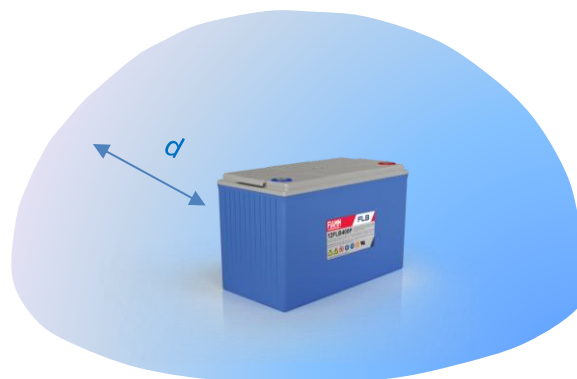


Fig. 13:
Proximity to the battery.

where N depends on the number of cells per monoblock battery (N) or vents openings per cell involved (1/N).

For further information please refer to IEC 62485-2 Standard or contact FET at: info.standby@fiamm.com

Note: A calculation program is available on request.